

Incident Handler's Journal

Google Cybersecurity Professional Certificate Portfolio

Incident Documentation | Entry #1 | May 19, 2026

Introduction

This journal documents security incidents investigated as part of the Google Cybersecurity Certificate program. Each entry applies the structured 5 W's framework to capture key incident intelligence in a consistent, professional format. This document serves as a portfolio artifact demonstrating practical incident response and documentation skills.

Incident Overview

A small U.S. healthcare clinic specializing in primary-care services experienced a severe ransomware attack on a Tuesday morning at approximately 9:00 a.m. The attack completely disrupted business operations — employees were unable to access patient medical records or the software required to perform their duties. A ransom note displayed on affected computers demanded a large sum of money in exchange for a decryption key to restore the encrypted files.

The attackers gained access through targeted phishing emails sent to multiple employees. The phishing emails contained malicious attachments that silently installed malware when opened. Once inside the network, the attackers deployed ransomware that encrypted all critical files. The organization was forced to shut down all computer systems and contact external organizations for technical assistance and incident reporting.

Journal Entry Details

Date: May 19, 2026

Entry #: 1

Severity: High

Description

A ransomware attack on a U.S. healthcare clinic led to complete operational shutdown. Phishing emails with malicious attachments served as the initial access vector. Ransomware was deployed across the network, encrypting patient records and critical operational files. A ransom demand was issued for a decryption key.

Tool(s) Used

No cybersecurity tools were actively deployed at the time of the attack. The following tools would be engaged during the incident response phase:

- Antivirus / Anti-malware scanner
- Endpoint Detection and Response (EDR) platform
- Network traffic analyzer (e.g., Wireshark or tcpdump)
- SIEM — Security Information and Event Management system
- Email security gateway (for phishing artifact analysis)
- Forensic imaging tool (e.g., FTK Imager) for evidence preservation

The 5 W's of the Incident

Who caused the incident?

An organized group of unethical hackers — a ransomware criminal organization known to deliberately target organizations in the healthcare and transportation industries. The group operates with technical sophistication and uses social engineering as an initial attack vector.

What happened?

Targeted phishing emails containing malicious attachments were sent to multiple clinic employees. When an employee downloaded the attachment, malware was silently installed on their computer. The attackers then deployed ransomware across the clinic's network, encrypting all critical files including patient medical records. A ransom note was displayed on all affected systems demanding payment for the decryption key, and all business operations halted.

When did the incident occur?

Tuesday morning at approximately 9:00 a.m. This timing is consistent with ransomware threat actor tactics — deploying attacks at the start of a business day maximizes disruption and increases the likelihood that employees will engage with email, making phishing more effective.

Where did the incident happen?

The incident originated on an employee workstation within the clinic's internal network via a phishing email attachment. Ransomware then propagated laterally across all connected systems, including servers storing patient medical records and operational software.

Why did the incident happen?

The incident succeeded due to three key failures: (1) Employees were susceptible to phishing, indicating gaps in security awareness training or email filtering. (2) Endpoint protection was insufficient to detect and block the malware before execution. (3) The absence of network segmentation and offline/immutable backups allowed the ransomware to spread freely and made recovery without paying the ransom difficult.

Additional Notes & Investigative Questions

This incident raises several important questions for further investigation:

- Was a ransom paid, and if so, was the decryption key successfully delivered?
- Were patient records (PHI) exfiltrated prior to encryption — indicating double extortion?
- Were offline or immutable backups available to restore systems without paying the ransom?
- How did the phishing email bypass the clinic's existing email security controls?
- What regulatory notifications are required under HIPAA given potential PHI exposure?

Recommended Immediate Actions:

- Isolate all affected systems to prevent further lateral spread
- Preserve forensic evidence before initiating recovery
- Notify CISA and HHS/OCR as required by HIPAA regulations
- Engage a qualified incident response firm
- Restore systems from verified clean backups if available
- Conduct phishing awareness training for all staff post-incident

Reflections & Key Learnings

This journal entry reinforced several core cybersecurity principles:

- **The human element is the weakest link.** Phishing remains the most prevalent initial access vector. A single employee action cascaded into a full organizational compromise.
- **Defense in depth is essential.** Multiple security layers — email filtering, endpoint protection, network segmentation, and offline backups — must all be in place simultaneously. A gap in any single layer should not lead to catastrophic failure.
- **Healthcare is a high-priority target.** Ransomware groups deliberately target healthcare because patient data is irreplaceable in the short term, increasing the likelihood of ransom payment.
- **Incident response planning saves time and money.** The absence of a tested Incident Response Plan (IRP) forced the clinic to shut down and seek external help, dramatically increasing response time and operational impact.